

Phase 6: Project Testing Phase

Project Name: **FitSense (FitTrack AI)** — API Verification, Test Execution Matrix & Results

6.1 Testing Strategy & Execution Framework

Testing for the FitSense backend API ensures that every endpoint processes requests reliably, enforces JWT security barriers, validates input schemas against boundary constraints, and integrates correctly with MongoDB and Google Gemini AI. Verification was conducted using black-box integration and system testing with Postman and VS Code Thunder Client, following a strict sequence: (1) Authentication APIs, (2) Workout Management CRUD APIs, and (3) AI Recommendation and Insight APIs.

6.2 Test Environment & Configuration Baselines

Execution Server

- Node.js (v16+) runtime
- Express.js application
- Base URL:
http://localhost:5000
- Started via: npm start

Data Tier & Security

- MongoDB (port 27017)
- Mongoose ODM schemas
- JWT Bearer header auth
- bcryptjs (10 salt rounds)

Test Tooling & Identity

- Thunder Client (VS Code)
- Postman client test suite
- Test User: Sathya
- Email: sathya@gmail.com

6.3 Comprehensive Test Execution Matrix (Part 1: Auth & Workouts)

Test Case / ID	Endpoint & Method	Test Input Payload / Headers	Expected Response Contract	Status
TC-01 User Register	POST /api/auth/register	JSON body: { "name": "Sathya", "email": "sathya@gmail.com", "password": "Sathya@01" }	HTTP 201 Created; returns signed JWT token string and sanitized user details (id, name, email).	PASS
TC-02 User Login	POST /api/auth/login	JSON body: { "email": "sathya@gmail.com", "password": "Sathya@01" }	HTTP 200 OK; verifies password via bcrypt and generates active JWT token.	PASS
TC-03 Get Profile	GET /api/auth/profile	Headers: Authorization: Bearer <token>; Body: None.	HTTP 200 OK; middleware decodes token and returns authenticated user details.	PASS
TC-04 Add Workout	POST /api/workouts/	Headers: Bearer <token> JSON: { "workoutName": "Evening Jog", "category": "Running", "duration": 30, "caloriesBurned": 350, "workoutDate": "2026-06-26" }	HTTP 201 Created; persists document in MongoDB with linked user ObjectId 6a40cc1091a5f2bf21887bc7.	PASS
TC-05 Get All Workouts	GET /api/workouts/	Headers: Bearer <token>; Body: None.	HTTP 200 OK; returns count and array of workouts owned by the authenticated user.	PASS

6.4 Observed Execution Performance (Auth & Initial Workouts)

Registration (TC-01) completed in **315 ms** (282 bytes), Login (TC-02) in **189 ms** (282 bytes), and Profile Retrieval (TC-03) in **75 ms** (100 bytes). Workout creation (TC-04) completed in **29 ms** (303 bytes), confirming fast local database transactions and minimal middleware overhead.

6.5 Comprehensive Test Execution Matrix (Part 2: Updates & AI Integration)

Test Case / ID	Endpoint & Method	Test Input Payload / Headers	Actual Output / Verification Data	Status
TC-06 Get Workout By ID	GET /api/workouts/:id	Headers: Bearer <token> Param: _id = 6a40cda891a5f2bf21887bcc	Status 200 OK (12 ms, 303 bytes); retrieved document matches specific ID.	PASS
TC-07 Update Workout	PUT /api/workouts/:id	Headers: Bearer <token> JSON: { "duration": 60, "caloriesBurned": 650 } with workout details	Status 200 OK (94 ms, 303 bytes); fields modified, timestamp updated.	PASS
TC-08 Delete Workout	DELETE /api/workouts/:id	Headers: Bearer <token> Param: target workout ID	Status 200 OK (69 ms, 57 bytes); returns "message": "Workout removed successfully".	PASS
TC-09 AI Recommendation	POST /api/ai/workout-rec	JSON: { "age": 22, "fitnessGoal": "Weight Loss", "experience": "Beginner" }	Status 200 OK (7.28 s, 1000 bytes); returns personalized multi-day routine with warm-up, core exercises, and safety advice.	PASS
TC-10 AI Fitness Insights	POST /api/ai/fitness-insights	JSON: { "totalWorkouts": 18, "averageDuration": 45, "totalCaloriesBurned": 6200 }	Status 200 OK (5.01 s, 275 bytes); returns analytical summary and progressive overload recommendations.	PASS

6.6 Testing Levels & Validation Categories

Functional & Integration Testing Authentication Flow: Verified registration hashing, JWT generation, and protected route access. Workout CRUD: Verified creation, retrieval, updates, and deletion of workout documents. Middleware Pass-Through: Verified that requests without a Bearer token are rejected with 401 Unauthorized. External API Services: Verified payload transmission to Google Gemini and parsing of the response text.	System Testing & UAT Scope End-to-End Execution: Validated the complete data pipeline from client request to database persistence and response delivery. Centralized Error Handling: Verified structured JSON error responses for invalid routes and payloads. User Acceptance Testing (UAT): <i>Not specified in the project document.</i> System testing was verified via automated and manual API client test suites.
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6.7 Issues, Bugs & Performance Findings

Bugs & Fixes: <i>Not specified in the project document</i> as having caused runtime crashes; endpoints operated with 100% schema conformance. AI Latency Profile: External AI endpoints exhibited response durations between 5.01 s and 7.28 s due to prompt evaluation by Google Gemini, confirming the necessity of keeping the AI layer as an isolated, asynchronous service.
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6.8 Documented Verification Artifacts & Test Recordings	
API Testing Video Recording	https://drive.google.com/file/d/1TCXG48a-78ryxF3gipdRMwRBwuhAgPaa/view?usp=drive_link
System Demo Video Walkthrough	https://drive.google.com/file/d/1KswrctSaPI0eOYyJ3GkbbVHvHsU3EfZJ/view?usp=sharing
Complete Project Drive Folder	https://drive.google.com/drive/folders/1hwmKxkHQkUI62JR1svI3xCQ2ShZ6Lj9y